

Prathyush Kumar Reddy Lebaku

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Summary

Analytical and adaptable Data Scientist with 3+ years of experience developing scalable ML pipelines, deploying deep learning models, and operationalizing solutions in real-time cloud environments. Skilled in Python, SQL, PySpark, AWS, and Azure with expertise in Computer Vision, NLP, LLMs, and MLOps. Adept at communicating complex insights, collaborating across teams, and delivering data-driven solutions that improve performance and accuracy.

Education & Certifications

University of Houston, Houston, TX

Master of Science in Data Science – Coursework: NLP, Data Analytics, AI/ML, Data Mining, Deep Learning

Aug 2023 – May 2025

CGPA: 3.91/4

Vellore Institute of Technology, Vellore, India

Bachelor of Technology in Computer Science and Engineering – Coursework: DSA, DBMS, Data Visualization

Aug 2019 – May 2023

CGPA: 3.62/4

Microsoft Power BI (PL-300), Advanced SQL (by Hacker Rank), Deep Learning (by Andrew Ng), Azure Data Scientist Associate, AWS Certified Cloud Practitioner

Technical Skills

Technical Skills: Python, SQL, PySpark, Scikit-learn, TensorFlow, PyTorch, Hugging Face, Flask, Streamlit, Git, AWS, Azure, GCP, Power BI, Tableau, Excel, NumPy, Pandas, OpenCV, NLTK, SpaCy

Soft Skills: Analytical Thinking, Communication, Collaboration, Problem-Solving, Adaptability, Attention to Detail

Professional Experience

Data Scientist | Computer Vision, Gen AI, LLMs

Jan 2024 – Present

University of Houston

Houston, TX.

- Built a real-time PPE violation detector using LLaMA-1.5v-13B, YOLO-World, and Haarcascade, improving safety compliance by 40%.
- Optimized road crack detection models (Faster R-CNN, YOLOv12) on CRDDC 2024 dataset, achieving up to 70% F1-score through ensembling.
- Created Bitcoin price predictor with PySpark + LSTM, reaching $R^2 = 0.89$ and cutting processing time by 65%.
- Developing a multimodal deep learning pipeline using CNN ensembles and Vision-Language Models (VLMs) to assess street-level walkability for urban planning; achieved ~90% accuracy with image classifiers and 85.4% with VLM-generated text classified by BERT, enabling scalable pedestrian environment evaluation.

Data Analyst | Machine Learning, Geospatial, Statistics, Tableau

Sep 2023 – Dec 2023

University of Houston affiliated with Iowa DOT

Houston, TX

- Merged 700K+ crash records and 4,119 road segments using GeoPandas, Shapely, and web-scraped data to generate spatial insights.
- Applied ANOVA and Tukey-Kramer tests to statistically assess pavement effects on crash rates; improved safety intervention accuracy by 35%.
- Created interactive Tableau dashboards and validated ML models with pseudo- $R^2 = 0.72$, supporting a \$200K grant.
- Presented research at the 2024 Cyber-Care Symposium, receiving recognition for analytical rigor and public safety impact.

Data Scientist Intern | PySpark, Azure, LSTM, Power BI

Jun 2022 – Jun 2023

Honeywell – Bangalore, India

- Engineered energy usage forecasting models using scikit-learn on 5TB+ telemetry data, improving forecast accuracy by 22%.
- Developed anomaly detection pipeline using LSTM and autoencoders, detecting faults with 93% precision.
- Created Power BI dashboards for real-time KPI monitoring; increased executive decision speed by 30%.
- Built a GPT-3-powered assistant for predictive maintenance Q&A and integrated it into production systems.
- Automated MLOps workflows with Azure ML, achieving 30% reduction in feature engineering time via reusable data pipelines.

Technical Projects

Talk to Your PDFs (Voice-Enabled RAG System with Chat history and Query Refinement) | Lang Chain

Mar 2025 – May 2025

- Created a multi-agent RAG pipeline with Streamlit UI and voice query for interactive search across 1,000+ PDFs with memory.
- Leveraged LangChain, Hugging Face, FAISS, and Mini-VLM, improving document retrieval and comprehension efficiency by 2x.

Satellite Imagery Pavement Condition Classification | Transfer Learning, Ensemble, AWS

Jan 2024 – May 2024

- Trained 11 transfer learning models on AWS SageMaker, ensemble-boosting accuracy from 73% to 91% and F1-score to 90%.
- Used early stopping, dropout, and regularization to prevent overfitting; deployed model to Amazon S3 for low-latency inference.
- Presented at AISIM-2024, winning 3rd prize out of 40+ participants.

Zillow House Value Prediction | SQL, Python (PySpark, Scikit-learn), Power BI, Excel

Aug 2024 – Dec 2024

- Processed 90,000+ property records using Python and SQL; engineered features revealing location-based price influencers.
- Trained Random Forest and XGBoost models reaching $R^2 = 0.88$; improved pricing workflow efficiency by 50% via Power BI dashboards.

Additional Projects:

- Sign Language Recognition using MediaPipe + LSTM, achieving 92% accuracy on 20+ gestures.
- QA Chatbot with BERT/BiDAF fine-tuning, enhancing response accuracy on custom domain FAQs.

Publications

- Evaluating the Impact of Pavement Conditions on Road Safety by Analyzing Crash Frequency and Severity (TRB 2025,USA)
- Deep Learning for Pavement Condition Evaluation Using Satellite Imagery. <https://doi.org/10.3390/infrastructures9090155> (MDPI)

Awards and Recognition

- Organized the Cyber-Care Challenge and developed RF and Encoder-Decoder based model to detect anomalies in **Connected Autonomous Vehicles**, improving safety through real-time anomaly. (uh.edu/cybercare/newsevents/cyber-care-competition/)
- Secured 3rd place among 40+ students for presenting Satellite Imagery Pavement Condition Classification results at AISIM-2024.
- Solved 150+ SQL problems in Leetcode and 100+ python problems.